

Gaussian Process-Based Model Predictive Control for Robust Autonomous Intersection Navigation Under Degraded V2I Communication

Daniel Szurek
Department of Computer Science
The University of Alabama
Tuscaloosa, AL, USA
djszurek@crimson.ua.edu

Chanakya Thirumala Setty
Department of Computer Science
The University of Alabama
Tuscaloosa, AL, USA
cthirumalasetty@crimson.ua.edu

Conner Hall
Department of Computer Science
The University of Alabama
Tuscaloosa, AL, USA
cchall5@crimson.ua.edu

Hwan-Sik Yoon
Department of Mechanical Engineering
The University of Alabama
Tuscaloosa, AL, USA
hyoon@eng.ua.edu

Paulius Puzinauskas
Department of Mechanical Engineering
The University of Alabama
Tuscaloosa, AL, USA
ppuzinauskas@eng.ua.edu

Abstract—This paper presents a Gaussian Process-based Model Predictive Control (GP-MPC) framework for Autonomous Intersection Navigation (AIN) that addresses robust eco-driving through signalized intersections under degraded Vehicle-to-Infrastructure (V2I) communication. A five-feature Automatic Relevance Determination (ARD) Gaussian Process predicts time-to-phase-change during V2I outages and outputs a predictive variance that quantifies confidence. This confidence signal directly modulates a state machine’s mode selection and a nonlinear MPC’s constraint conservatism, creating a unified uncertainty-aware control pipeline. Validation across 35 test configurations (7 scenarios $\times$ 5 noise profiles) demonstrates zero red-light violations under all conditions, including up to 33.1% packet loss. The comfort index (ISO 2631-1) degrades only 13.7% from ideal to severe noise, while average velocity remains within 1.1% of baseline. Correlation analysis confirms that comfort degrades with packet loss ($r = +0.676$, $p < 0.001$) while velocity tracking remains statistically invariant ($r = -0.065$, $p = 0.711$).

Index Terms—autonomous intersection navigation, Gaussian process, model predictive control, V2I communication, eco-driving, connected vehicles

I. INTRODUCTION

Connected and Automated Vehicles (CAVs) can exploit Vehicle-to-Infrastructure (V2I) communication for energy-efficient intersection navigation [1]–[3]. Signal Phase and Timing (SPaT) information enables predictive speed optimization that reduces unnecessary braking cycles. However, V2I reliability remains a critical challenge—packet loss rates of 10–30% are common beyond 100 m under IEEE 802.11p [11], and multi-second blackouts occur during infrastructure handoffs. Existing approaches largely assume reliable communication, leaving a gap in robust intersection control under degraded conditions.

This paper addresses this gap through a Gaussian Process (GP)-based Model Predictive Control (MPC) framework with

three complementary mechanisms: (1) a GP that predicts time-to-phase-change during communication loss and outputs a confidence measure, (2) a nonlinear MPC with intersection-aware constraints whose conservatism adapts to GP confidence, and (3) a state machine with phase filtering and committed-stop latching for safety. The novelty lies in using GP-derived predictive variance as the connective signal that bridges the state machine’s mode selection and the MPC’s constraint tightness, creating an end-to-end uncertainty-aware pipeline.

The primary contributions are:

- A five-feature Automatic Relevance Determination (ARD) GP for signal timing prediction with confidence output that modulates downstream control decisions.
- A novel integration architecture where GP uncertainty bridges a state machine and nonlinear MPC for end-to-end uncertainty-aware control.
- Validation across 35 configurations referenced to international Vehicle-to-Everything (V2X) standards, demonstrating zero safety violations with graceful degradation.

II. RELATED WORK

Intersection navigation for connected vehicles has been studied through two primary paradigms. System-centric approaches [4]–[6] propose centralized intersection management that coordinates vehicle trajectories, eliminating traffic signals entirely. While these demonstrate significant throughput improvements, they require all vehicles to be connected and cooperative—a prerequisite that limits near-term deployment.

Ego-centric approaches optimize a single vehicle’s trajectory using available SPaT data. Asadi and Vahidi [1] demonstrated predictive cruise control using upcoming signal information, achieving 10–15% fuel economy improvements.

Kamal et al. [2] extended this to MPC-based urban driving with velocity and position optimization. Yang et al. [3] incorporated queue effects into an eco-cooperative adaptive cruise control framework, and He et al. [8] formulated optimal speed trajectory planning on signalized arterials with queue consideration.

A common limitation of existing ego-centric methods is the assumption of reliable V2I communication. When SPaT data is lost or corrupted, these systems either halt (overly conservative) or continue with stale data (potentially unsafe). Zhang et al. [9] proposed virtual intersection models for robustness but did not explicitly quantify communication uncertainty.

State machines and MPC controllers are individually well-established in automated driving. State machines provide interpretable, verifiable mode logic for safety-critical decisions [2], while MPC provides optimal trajectory planning with constraint satisfaction [3]. However, neither alone addresses V2I uncertainty; a state machine without uncertainty information cannot distinguish reliable from unreliable SPaT data, and an MPC without adaptive constraints cannot modulate conservatism based on communication quality.

Our approach introduces a Gaussian Process as the connective element between these components. The GP provides both a prediction of time-to-phase-change and a confidence measure (predictive variance) that serves a dual role, informing the state machine's mode selection and modulating the MPC's constraint tightness. This creates a principled, continuous transition between full-information optimal control and uncertainty-aware degraded operation. Unlike reinforcement learning approaches [7], this framework provides interpretable uncertainty bounds and hard safety constraints, with the GP variance offering a physically meaningful measure of prediction reliability that decays predictably with communication outage duration.

III. SYSTEM ARCHITECTURE

The Autonomous Intersection Navigation (AIN) system follows a layered pipeline:

$$\begin{aligned} \text{V2I} &\rightarrow \text{Validity Monitor} \rightarrow \text{GP Switch} \\ &\rightarrow \text{State Machine} \rightarrow \text{MPC} \rightarrow a_{\text{cmd}} \end{aligned} \quad (1)$$

where a_{cmd} is the acceleration command output. All components operate at 10 Hz ($T_s = 0.1$ s). The system processes V2I data for up to three sequential intersections, each providing distance to stop line (d_{int}), signal phase ($\phi \in \{3, 6, 8\}$ for red, green, yellow), and time to phase change (t_{change} in deciseconds). Communication uses a User Datagram Protocol (UDP) at 10 Hz carrying 5 doubles (40 bytes).

The validity monitor detects communication degradation by tracking message timestamps and applying a staleness threshold. It latches the last valid t_{change} value for downstream use and outputs a boolean `is_valid` flag. The GP switch routes data based on communication status:

$$t_{\text{change}}^{\text{out}} = \begin{cases} t_{\text{change}}^{\text{raw}} & \text{if } \text{is_valid} = \text{true} \\ t_{\text{change}}^{\text{GP}} & \text{if } \text{is_valid} = \text{false} \end{cases} \quad (2)$$

IV. GAUSSIAN PROCESS MODEL

When V2I is lost, ϕ can be held at its last value and d_{int} can be dead-reckoned, but t_{change} cannot be extrapolated because phase transitions may have occurred during the blackout. The GP predicts t_{change} from five features available during blackout:

$$\mathbf{z} = [v_{\text{ego}}, d_{\text{int}}^{\text{dr}}, \phi_{\text{last}}, t_{\text{ds,last}}, \Delta t]^T \quad (3)$$

where v_{ego} is ego velocity (from Controller Area Network (CAN)), $d_{\text{int}}^{\text{dr}}$ is dead-reckoned distance, ϕ_{last} is last confirmed phase, $t_{\text{ds,last}}$ is last known time-to-phase-change, and Δt is elapsed time since last valid update. This extends the three-feature formulation of [1] by incorporating dead-reckoned distance and blackout duration, allowing the kernel to learn that predictions degrade with time since last contact.

The GP employs an ARD squared exponential kernel:

$$k(\mathbf{z}, \mathbf{z}') = \sigma_f^2 \exp \left(-\frac{1}{2} \sum_{i=1}^5 \frac{(z_i - z'_i)^2}{\ell_i^2} \right) \quad (4)$$

where per-feature length scales ℓ_i automatically identify the most informative features. A constant basis function captures the mean offset.

Critically, the GP outputs both a mean prediction μ and predictive variance σ_{pred}^2 serving as the system's *confidence signal*. The variance grows from the GP's intrinsic variance as query points move from training data, and an explicit temporal growth model:

$$\sigma_{\text{pred}}^2(t) = \begin{cases} \sigma_{\text{base}}^2 & \text{if V2I active} \\ \sigma_{\text{base}}^2 \cdot (1 + \beta \cdot \Delta t) & \text{if V2I lost} \end{cases} \quad (5)$$

This ensures confidence degrades monotonically with blackout duration. The confidence signal governs three operating regimes:

- **High confidence** ($\sigma_{\text{pred}}^2 \leq 2.0$): GP prediction fully trusted for Green Light Optimal Speed Advisory (GLOSA) and cruise decisions.
- **Moderate confidence** ($2.0 < \sigma_{\text{pred}}^2 \leq 5.0$): Prediction used with increased safety margins.
- **Low confidence** ($\sigma_{\text{pred}}^2 > 5.0$): Conservative Stop forced regardless of predicted phase.

V. STATE MACHINE

The state machine translates GP-augmented V2I data into MPC reference signals: velocity reference (v_{ref}), distance constraint (d_{safe}), and operating mode. Table I summarizes the three operating modes.

TABLE I: State Machine Operating Modes

Mode	Name	v_{ref}	d_{safe}
1	Cruise	v_{set}	∞ (no constraint)
2	Stop	Braking profile	d_{int}
3	GLOSA	Advisory speed	d_{int}

A phase filter requires $N_{\text{confirm}} = 3$ consecutive identical readings (0.3 s) before accepting a phase change, preventing

single-sample corruption from triggering inappropriate mode switches.

The decision logic follows strict priority: (1) past intersection $\rightarrow$ Cruise; (2) confirmed GREEN $\rightarrow$ Cruise (precedes uncertainty checks to prevent `is_valid` flickering from injecting stops during green); (3) committed stop latch—if previously committed within 1.5 m, hold Stop until GREEN (prevents noise-induced creep); (4) far from intersection (estimated time of arrival (ETA) ≥ 10 s) $\rightarrow$ Cruise; (5) high GP uncertainty on RED/YELLOW ($\sigma_{\text{pred}}^2 \geq 5.0$, V2I invalid) $\rightarrow$ Stop; (6) YELLOW: evaluate clearance feasibility; (7) RED: evaluate GLOSA feasibility.

In Stop mode, the reference follows a kinematic braking envelope:

$$v_{\text{brake}} = \sqrt{2 \cdot a_{\text{target}} \cdot \max(0, d_{\text{int}} - d_{\text{buf}})} \quad (6)$$

where $a_{\text{target}} = 1.5 \text{ m/s}^2$ and $d_{\text{buf}} = 0.5 \text{ m}$. For GLOSA, the advisory speed is $v_{\text{GLOSA}} = d_{\text{int}} / (t_{\text{green}} + 2.5)$, activated only when within 5.0 m/s of set speed.

The state machine maintains dead-reckoned distance during blackouts: $d_{\text{int}}^{\text{dr}}(k+1) = d_{\text{int}}^{\text{dr}}(k) - v_{\text{ego}}(k) \cdot T_s$, reset when V2I recovers.

VI. MPC FORMULATION

The nonlinear MPC uses a two-state longitudinal model:

$$p_{k+1} = p_k + v_k \cdot T_s + \frac{1}{2} a_k \cdot T_s^2 \quad (7)$$

$$v_{k+1} = v_k + a_k \cdot T_s \quad (8)$$

As shown in Table II, the high output weight ($w_y = 2.0$) ensures aggressive velocity tracking, the low input weight ($w_u = 0.05$) allows sustained braking for red-light stops, and the moderate rate weight ($w_{\Delta u} = 0.8$) penalizes jerk for comfort.

TABLE II: MPC Configuration Parameters

Parameter	Symbol	Value
Prediction / Control horizon	N_p / N_c	30 / 10
Output / Input / Rate weight	$w_y / w_u / w_{\Delta u}$	2.0 / 0.05 / 0.8
Accel. limits	$a_{\text{min}}, a_{\text{max}}$	$\pm 3.0 \text{ m/s}^2$
Velocity limits	$v_{\text{min}}, v_{\text{max}}$	0, 20 m/s

The MPC enforces a custom intersection-aware inequality constraint:

$$v_k \leq \sqrt{2 \cdot a_{\text{brake}} \cdot \max(0, d_{\text{safe}} - p_k - d_{\text{margin}})} \quad (9)$$

where $a_{\text{brake}} = 2.5 \text{ m/s}^2$ and $d_{\text{margin}} = 5.0 \text{ m}$. This dual-layer design—comfortable reference at 1.5 m/s^2 from the state machine, hard constraint at 2.5 m/s^2 from the MPC—allows smooth approaches while guaranteeing the safety limit is never violated.

The cost function minimizes:

$$J = \sum_{k=0}^{N_p-1} [w_y(v_k - v_{\text{ref}})^2 + w_u a_k^2 + w_{\Delta u}(\Delta a_k)^2] \quad (10)$$

VII. VALIDATION FRAMEWORK

A. Simulation Environment

The framework is validated using a Python simulation providing standards-referenced V2I noise modeling using the same UDP protocol used by the MATLAB/Simulink harness. Each run produces time-series data (15 channels at 10 Hz) and 17 performance metrics across safety, efficiency, comfort, energy, communication quality, and velocity tracking.

B. Test Scenarios

Table III lists the seven scenarios covering the operational design domain.

TABLE III: Test Scenario Specifications

Scenario	Description	Int.	BO
Red Light Stop	Approach red, decelerate and stop	2	0
GLOSA Advisory	Red turning green, optimize arrival	2	0
Green Cruise	Green ahead, cruise through	2	0
Dilemma Zone	Yellow at 120 m, commit or stop	2	0
V2I Blackout	8 s V2I loss mid-approach	2	1
Urban Corridor	5 intersections, mixed phases	5	0
GP Stress Test	Multiple blackouts, max GP reliance	3	3

Int. = intersections; BO = blackout windows.

C. V2I Noise Profiles

Table IV summarizes five noise profiles that model communication degradation at increasing severity.

TABLE IV: V2I Communication Noise Profiles

Profile	σ_d (m)	σ_t (s)	Phase Err.	Pkt. Loss	Standard
Ideal	0.0	0.0	0.0%	0.0%	Baseline
ETSI Nom.	1.5	0.2	0.5%	2.0%	ETSI ITS-G5
SAE Urban	3.0	0.8	2.0%	10.0%	SAE J2945/1
IEEE Deg.	5.0	2.0	5.0%	20.0%	IEEE 802.11p
Severe	8.0	3.0	10.0%	30.0%	Stress test

Each profile models six noise dimensions: position noise (σ_d , Global Navigation Satellite System (GNSS) multipath [13]), timing noise (σ_t , Road-Side Unit (RSU) synchronization jitter [11]), phase error rate (bit errors at given Signal-to-Noise Ratio (SNR) [14]), packet loss (Packet Delivery Ratio (PDR) vs. range [12], [14]), stale data probability (congestion-induced delays [12]), and timestamp jitter [11].

VIII. RESULTS AND ANALYSIS

All 35 test configurations used a set speed of 20 m/s, initial speed of 15 m/s, and $T_s = 0.1 \text{ s}$.

A. Safety

Zero red-light violations were recorded across all 35 runs, including the GP Stress Test with Severe noise (33.1% effective packet loss). The minimum recorded stop distance was 5.00 m with a mean of 26.17 m. The committed-stop latch proved essential. In the Dilemma Zone with Severe noise, distance fluctuations of $\pm 16 \text{ m}$ (2σ at $\sigma_d = 8.0 \text{ m}$) could produce non-zero references while stopped, but the latch maintained the stop.

B. Comfort and Efficiency

Table V presents comfort, jerk, velocity, and energy across noise profiles. The comfort index increases by only 13.7% from Ideal to Severe, with all values within the ISO 2631-1 “a little uncomfortable” range (0.5–1.63). Notably, the comfort index saturates between IEEE Degraded and Severe, indicating the phase filter effectively bounds noise impact beyond edge-of-coverage conditions. Average velocity decreases by only 1.1% (12.32 to 12.18 m/s), demonstrating preserved throughput. Energy increases 7.0%, reflecting corrective accelerations under noise.

TABLE V: Performance by Noise Profile (Mean $\pm$ Std)

Noise Profile	Comfort	RMS Jerk	$\bar{v}$ (m/s)	Energy (J/m)
Ideal	1.43 $\pm$ 0.29	2.34 $\pm$ 0.44	12.32 $\pm$ 2.61	0.402 $\pm$ 0.13
ETSI Nom.	1.48 $\pm$ 0.28	4.44 $\pm$ 0.70	12.29 $\pm$ 2.60	0.409 $\pm$ 0.13
SAE Urban	1.55 $\pm$ 0.28	5.87 $\pm$ 1.11	12.26 $\pm$ 2.61	0.413 $\pm$ 0.13
IEEE Deg.	1.62 $\pm$ 0.26	6.80 $\pm$ 0.85	12.25 $\pm$ 2.59	0.424 $\pm$ 0.13
Severe	1.63 $\pm$ 0.28	6.78 $\pm$ 1.05	12.18 $\pm$ 2.57	0.430 $\pm$ 0.13

C. Per-Scenario Analysis

Table VI presents per-scenario results. The GLOSA Advisory achieves the best energy efficiency (0.273 J/m)—a 38.4% reduction vs. the mean—confirming the value of predictive speed optimization. The GP Stress Test produces the highest comfort index (2.024) due to 21 s of total blackout, but maintains zero violations. The Urban Corridor demonstrates effective multi-intersection coordination with low comfort index (1.329) and energy (0.304 J/m) across five intersections.

TABLE VI: Per-Scenario Performance (Averaged Across Noise Profiles)

Scenario	$\bar{v}$ (m/s)	Comfort	Energy	Vel. RMSE
Red Light Stop	14.36	1.627	0.443	8.983
GLOSA Advisory	14.37	1.244	0.273	9.708
Green Cruise	12.39	1.388	0.338	11.340
Dilemma Zone	7.04	1.442	0.459	15.041
V2I Blackout	13.74	1.717	0.448	9.892
Urban Corridor	12.98	1.329	0.304	10.721
GP Stress Test	10.97	2.024	0.643	12.103

D. Communication Quality Impact

Table VII shows the Pearson correlation results. A clear dichotomy emerges; comfort and energy are significantly correlated with packet loss, while velocity metrics are statistically invariant. The GP-MPC preserves trajectory-level performance regardless of communication quality, incurring higher-frequency control effort under degraded conditions—the expected behavior of a well-designed robust controller.

TABLE VII: Pearson Correlation: Packet Loss vs. Performance

Metric	r	p -value
Comfort Index (ISO 2631-1)	+0.676	< 0.001
RMS Jerk (m/s ³)	+0.683	< 0.001
Energy Proxy (J/m)	+0.485	0.003
Avg. Velocity (m/s)	−0.065	0.711
Velocity RMSE (m/s)	+0.060	0.732

E. Degradation Analysis

Table VIII quantifies the Ideal-to-Severe change per scenario. The V2I Blackout shows the largest comfort degradation (+0.464) because the 8-second blackout overlaps with critical braking, compounding GP uncertainty with noisy recovery signals. The GP Stress Test shows minimal degradation (+0.052) since it already operates under heavy uncertainty in the Ideal profile. Negative energy deltas for Red Light Stop and Dilemma Zone indicate that earlier uncertainty-triggered braking reduces peak kinetic energy that must be dissipated.

TABLE VIII: Performance Degradation: Ideal $\rightarrow$ Severe

Scenario	Δ Comfort	Δ Vel (m/s)	Δ Energy (J/m)
Red Light Stop	+0.133	+0.36	−0.036
GLOSA Advisory	+0.116	+0.00	+0.018
Green Cruise	+0.246	+0.17	+0.025
Dilemma Zone	+0.166	+0.06	−0.070
V2I Blackout	+0.464	+0.22	+0.168
Urban Corridor	+0.195	+0.04	+0.027
GP Stress Test	+0.052	+0.11	+0.062

IX. DISCUSSION

The validation supports three primary conclusions. First, safety is maintained under all conditions; zero violations across 35 runs validate the defense-in-depth approach combining phase filtering, committed-stop latching, and MPC kinematic constraints. Second, degradation is graceful and bounded; comfort saturates between IEEE Degraded and Severe, while velocity is statistically invariant to noise. Third, GP confidence enables principled degradation; the correlation between packet loss and comfort ($r = +0.676$) but not velocity ($r = -0.065$) demonstrates that macro-level optimality is maintained while accepting micro-level smoothness degradation—exactly the trade-off encoded by the GP variance thresholds.

Key design trade-offs include the phase filter depth ($N_{\text{confirm}} = 3$, 0.3 s latency vs. noise rejection), the dual-layer braking (comfortable 1.5 m/s² reference vs. hard 2.5 m/s² constraint), and explicit Δt in the GP features (supplements linear variance growth with learned non-linear staleness effects).

Limitations include longitudinal-only control, simplified two-state dynamics, fixed offline GP, and single-vehicle evaluation without surrounding traffic. The observed RMS jerk increase (190% Ideal to Severe) represents raw MPC output; in a full Advanced Driver Assistance System (ADAS) integration, downstream rate limiters would substantially attenuate these transients.

X. CONCLUSION

This paper presented a GP-MPC framework for Autonomous Intersection Navigation that addresses V2I communication uncertainty through a novel integration of a Gaussian Process, state machine, and nonlinear MPC. The GP's predictive variance serves as a confidence signal bridging mode selection and constraint adaptation, creating a unified uncertainty-aware pipeline.

Validation across 35 configurations demonstrated zero red-light violations under all conditions (up to 33.1% packet loss, 21 s blackout), 13.7% comfort degradation from ideal to severe conditions, velocity statistically invariant to communication quality, and GLOSA advisory achieving 38.4% energy reduction.

Future work includes online GP adaptation for intersection-specific patterns, integration with full ADAS rate limiting for jerk reduction, lateral path planning coupling [10], and powertrain-aware MPC with regenerative braking optimization.

ACKNOWLEDGMENT

This research is conducted as part of the EcoCAR EV Challenge, organized by Argonne National Laboratory. The authors thank the competition organizers and sponsors for their support.

REFERENCES

- [1] B. Asadi and A. Vahidi, "Predictive cruise control: Utilizing upcoming traffic signal information for improving fuel economy and reducing trip time," *IEEE Trans. Control Syst. Technol.*, vol. 19, no. 3, pp. 707–714, 2011.
- [2] M. A. S. Kamal, M. Mukai, J. Murata, and T. Kawabe, "Model predictive control of vehicles on urban roads for improved fuel economy," *IEEE Trans. Control Syst. Technol.*, vol. 21, no. 3, pp. 831–841, 2013.
- [3] H. Yang, H. Rakha, and M. V. Ala, "Eco-cooperative adaptive cruise control at signalized intersections considering queue effects," *IEEE Trans. Intell. Transport. Syst.*, vol. 18, no. 6, pp. 1575–1585, 2017.
- [4] M. Hausknecht, T.-C. Au, and P. Stone, "Autonomous intersection management: Multi-intersection optimization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 4581–4586.
- [5] D. Fajardo, T.-C. Au, S. T. Waller, P. Stone, and D. Yang, "Automated intersection control: Performance of future innovation versus current traffic signal control," *Transp. Res. Rec.*, vol. 2259, no. 1, pp. 223–232, 2011.
- [6] A. Parker and G. Nitschke, "How to best automate intersection management," in *Proc. IEEE Congr. Evol. Comput.*, 2017, pp. 1247–1254.
- [7] Q. Sun et al., "Hierarchical reinforcement learning for dynamic autonomous vehicle navigation at intelligent intersections," in *Proc. ACM SIGKDD*, 2023, pp. 4852–4861.
- [8] X. He, H. X. Liu, and X. Liu, "Optimal vehicle speed trajectory on a signalized arterial with consideration of queue," *Transp. Res. Part C*, vol. 61, pp. 106–120, 2015.
- [9] T. Zhang, D. Yang, T. Li, K. Li, and X. Lian, "An improved virtual intersection model for vehicle navigation at intersections," *Transp. Res. Part C*, vol. 19, no. 3, pp. 413–423, 2011.
- [10] L. Niklasson, "Intersection navigation for automated vehicles," M.S. thesis, Chalmers Univ. Technol., 2018.
- [11] SAE International, "V2X communications message set dictionary," SAE Standard J2735, 2024.
- [12] SAE International, "On-board system requirements for V2V safety communications," SAE Standard J2945/1, 2020.
- [13] ETSI, "Intelligent transport systems; Communication congestion control," ETSI TS 102 894-2, 2018.
- [14] IEEE, "Wireless access in vehicular environments," IEEE Std 802.11p-2010, 2010.
- [15] ISO, "Mechanical vibration and shock—Evaluation of human exposure to whole-body vibration," ISO 2631-1:1997, 1997.